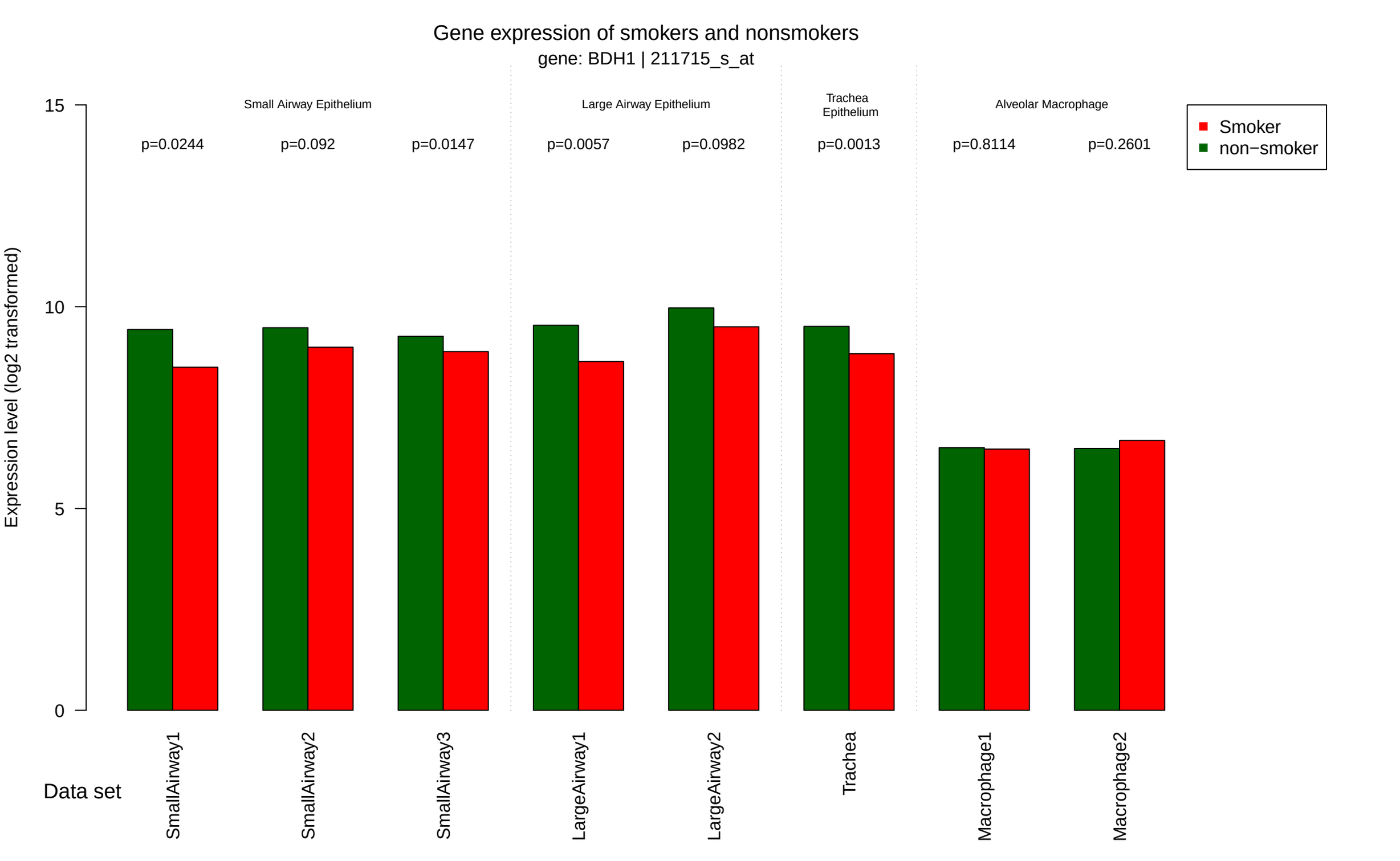
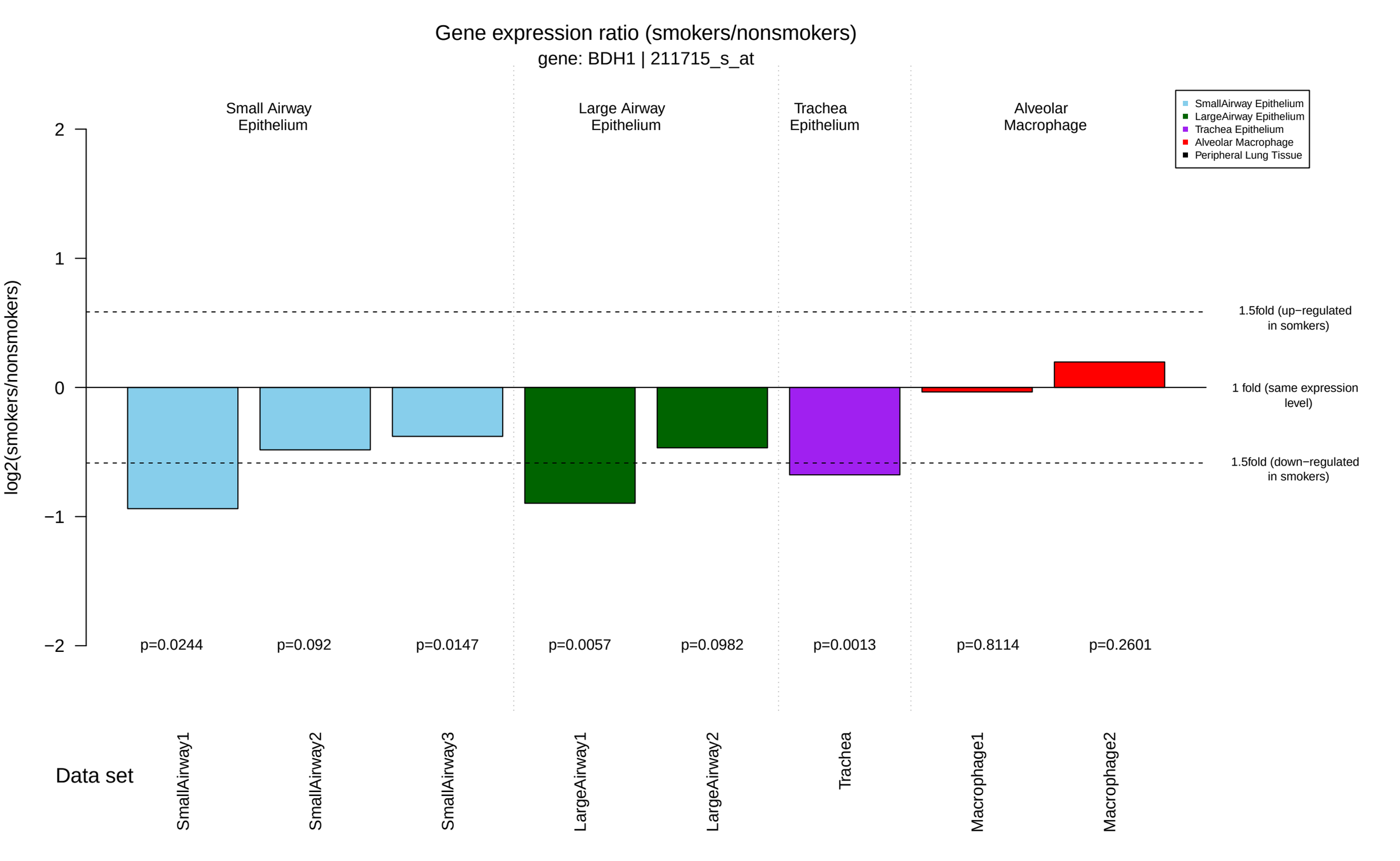
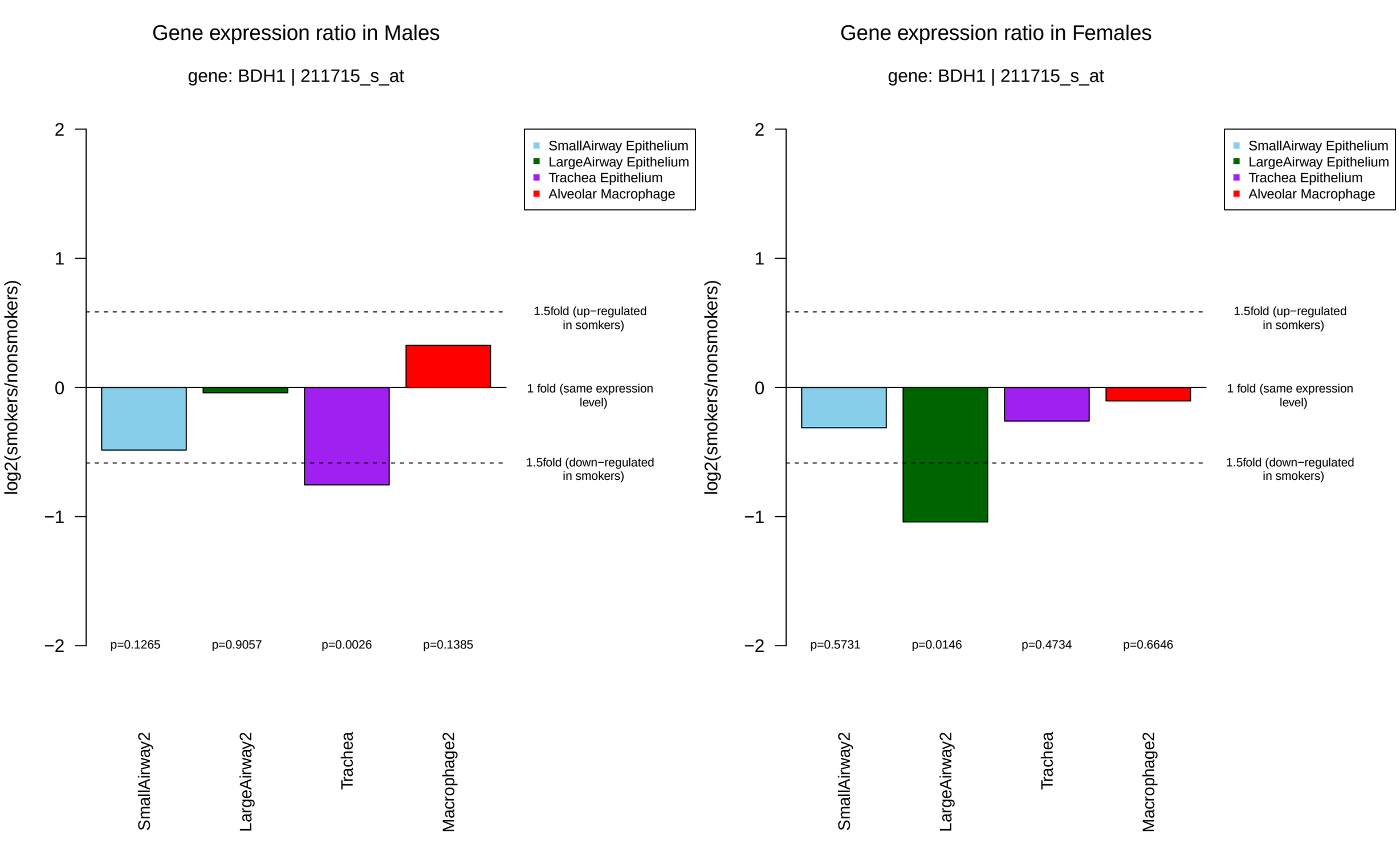
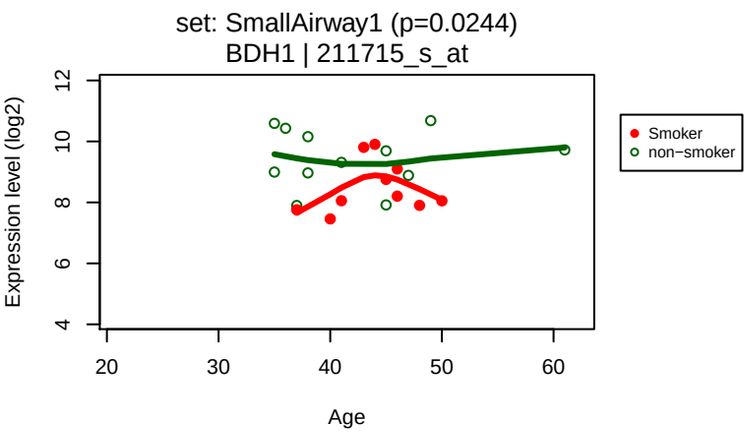
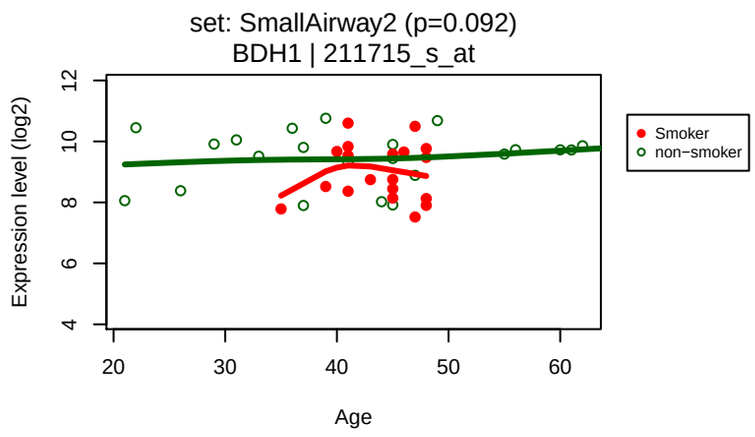
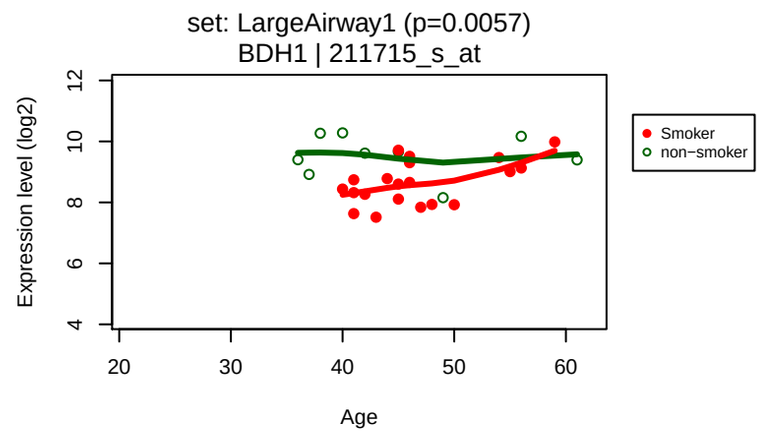

Gene expression of smokers and nonsmokers
gene: BDH1 | 211715_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0244
p=0.092
p=0.0147
p=0.0057
p=0.0982
p=0.0013
p=0.8114
p=0.2601
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

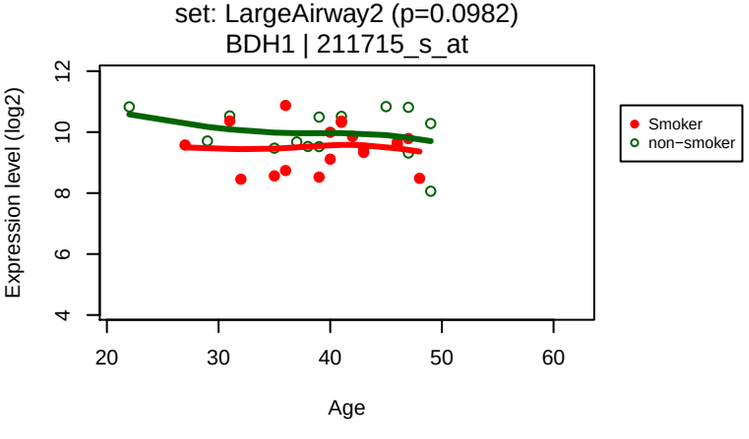

Gene expression ratio (smokers/nonsmokers)
gene: BDH1 | 211715_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0244
p=0.092
p=0.0147
p=0.0057
p=0.0982
p=0.0013
p=0.8114
p=0.2601
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

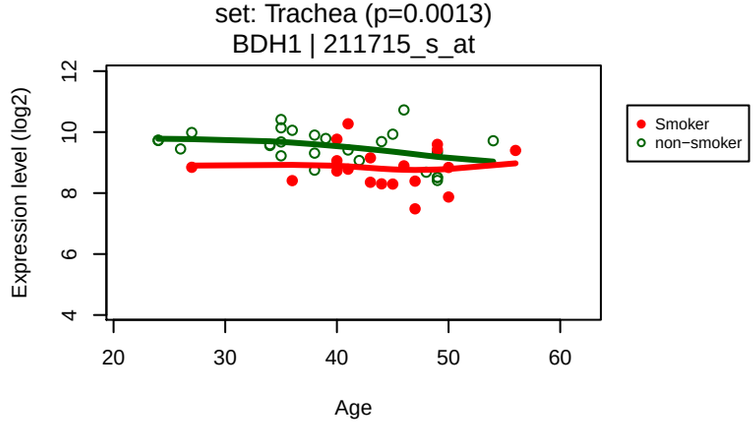

Gene expression ratio in Males
gene: BDH1 | 211715_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1265
p=0.9057
p=0.0026
p=0.1385
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BDH1 | 211715_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5731
p=0.0146
p=0.4734
p=0.6646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

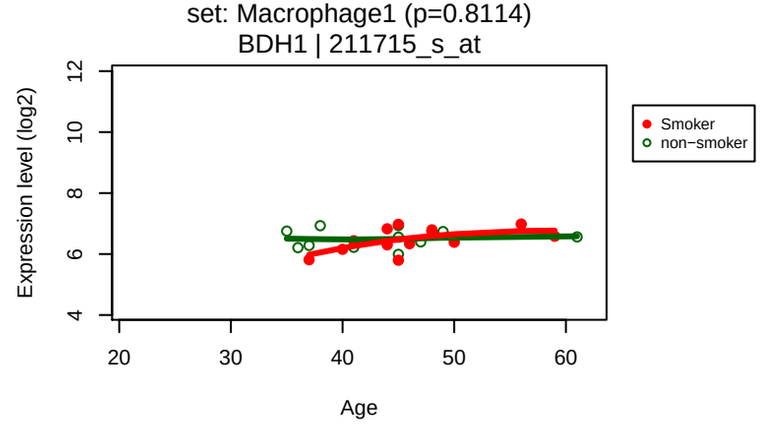

set: SmallAirway1 (p=0.0244)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker

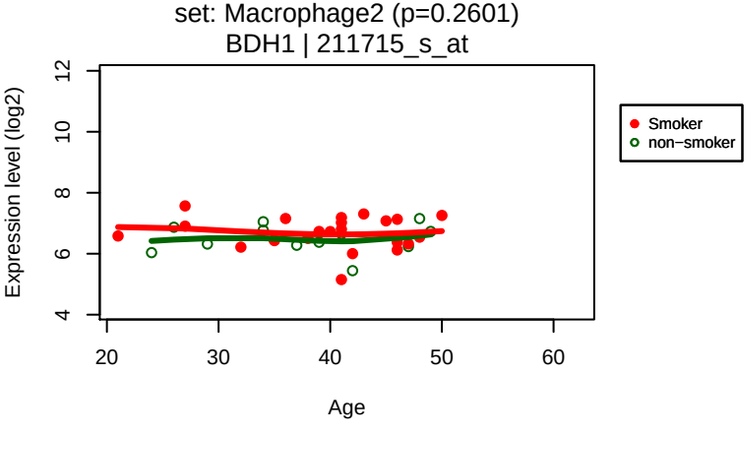

set: SmallAirway2 (p=0.092)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker

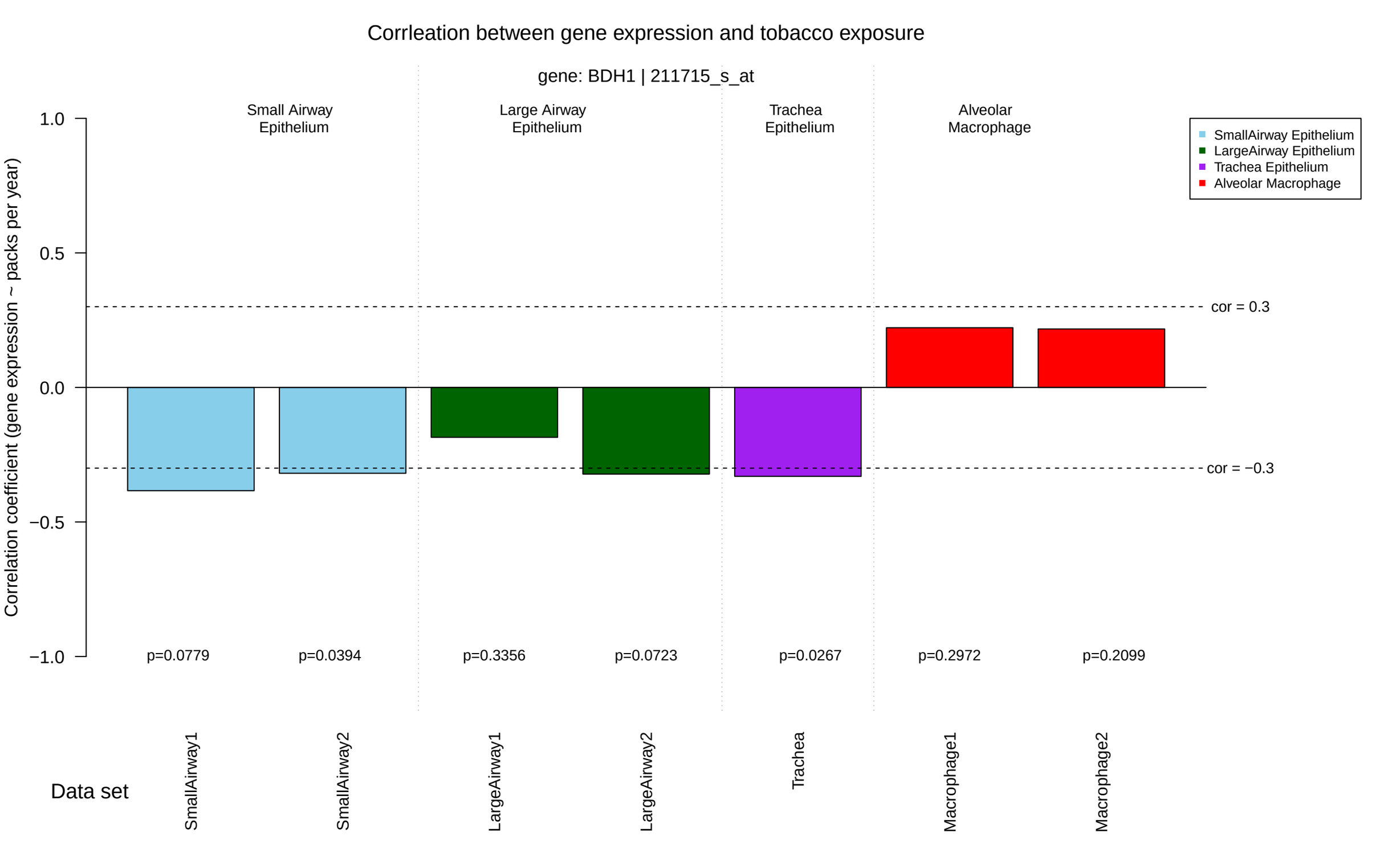

set: LargeAirway1 (p=0.0057)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0982)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8114)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2601)
BDH1 | 211715_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BDH1 | 211715_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0779
p=0.0394
p=0.3356
p=0.0723
p=0.0267
p=0.2972
p=0.2099
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2